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GEOG 489: Mapping with Drones

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Elevation and Vegetation Health Spatial Analysis of a Creswell, Oregon Hazelnut Orchard

Background

The Farmhands Co-op is a worker-owned cooperative located in Creswell, Oregon, that supports local agriculture through organic orchard practices, agroforestry, and community-centered land stewardship. Although the farm is diverse with several agricultural products, our analysis focuses primarily on the hazelnut orchard. Hazelnut tree health is influenced by many different environmental conditions. Through the use of drone imagery and LiDAR data, we analyze the ways in which these conditions affect hazelnut trees in this orchard.

Project Summary

On May 23rd, 2026, our class visited the Co-op in order to collect aerial data of a hazelnut orchard to analyze and answer our research questions. During the field day, we used a DJI Mavic drone to collect RGB (red, green, and blue bands) and multispectral (red, green, near-infrared, and red-edge bands) imagery and the DJI Matrice drone to collect LiDAR data. Our analysis was done using the coarse imagery, as it was sufficient to gather enough detail and improve processing times. In this project, our goal was to use the data to examine how hazelnut tree health varies spatially with a Normalized Difference Vegetation Index (NDVI) and a

Normalized Difference Red Edge Index (NDRE), as well as to identify topographic variations in the land and how they might affect drainage patterns or tree health.

To better understand how the data can be used to assess hazelnut health, we needed to identify the conditions that affect overall growth and health. Hazelnut groves do best in full sun (6+ hours); they have relatively shallow root systems, and healthy trees have a lifespan of approximately 40-80 years *(The Spruce)*. Hazelnut trees thrive in well-drained, slightly acidic soil—so having flatter areas or gentle slopes is best to prevent erosion and weak rooting. Our LiDAR analysis aimed to find areas with topographic variations that may inform about drainage patterns in the grove. The multispectral imagery was used to analyze canopy health and identify areas with greater coverage.

Methods

Data collection

Data was collected using two drones, a DJI Mavic and a DJI Matrice. Ground control points (GCPs) were placed, and an Emlid base station and rover were used to get the coordinates of these points. The DJI Mavic was flown over the hazelnut orchard at the Farmhands Co-op, collecting RGB and multispectral data. The DJI Matrice flew on a similar path, collecting LiDAR data. When collecting the LiDAR data, it was sunny with minimal wind, but when collecting the RGB and multispectral data, it was much windier, almost reaching the limits of the drone's capability. The LiDAR data were collected around 12 pm, and the multispectral and RGB data were collected much later in the day, a few days later. The multispectral and RGB data were collected at two resolutions, coarse and high. The coarse resolution had a GSD of around 400

feet, yielding a 5 cm/pixel. The high resolution has a GSD closer to 200 ft, yielding 2.5 cm/pixel. For our analysis, we used the coarse resolution to decrease processing time and energy.

Data Analysis

The data was processed in Agisoft Metashape Pro and ArcGIS Pro. GCP coordinates collected by the Emlid rover were used to georeference our data. First, an orthoimage and digital surface model (DSM) were created in Agisoft using the multispectral imagery. From this, we created an NDVI and an NDRE using the raster calculator in Agisoft. The NDVI is valuable for assessing vegetation health and density and is sensitive to early growth stages, and thus is typically used for early to mid-season crop monitoring. It uses the red and near-infrared (NIR) bands in the following equation:

$(NIR - Red)/(NIR + Red)$.

The NDRE uses the red edge and NIR bands. It provides insights into chlorophyll content and therefore measures stress for mature crops. The equation for the NDRE is shown below:

$(NIR - Red\,Edge)/(NIR + Red\,Edge)$

Next, we filtered the LiDAR point clouds (using a point spacing of 0.3 meters) and then cropped them to the extent of the orchard in Agisoft to decrease processing time. From the resulting point cloud, we created a DSM in ArcGIS Pro using the LAS to Raster tool and a cell size of 1 meter.

Results



Figure 1: Orthoimage of the main hazelnut orchard at the Farmhands Co-op created using Agisoft Metashape Pro.

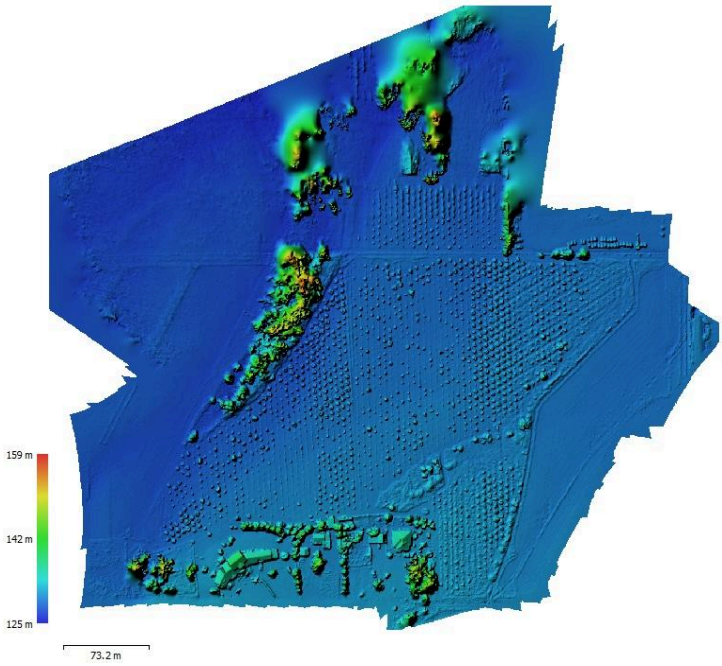


Figure 2: DSM of the main hazelnut orchard at the Farmhands Co-op created using Agisoft Metashape Pro.

The tallest trees border the northwest section of the hazelnut orchard. South of the orchard, houses and other tree species are represented on the DSM.

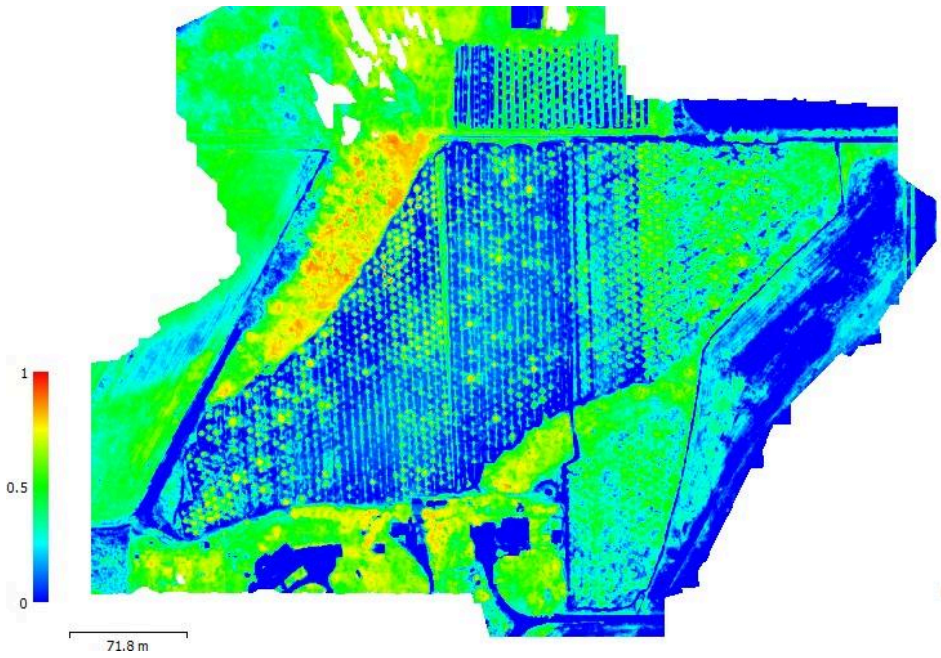


Figure 3: NDVI of the main hazelnut orchard created on Agisoft Metashape Pro.

NDVI values lower than 0.1 indicate bare rock, sand, and snow. In the southernmost part of the orchard, there are houses with NDVI values of 0.1 or less. NDVI values of 0.2 - 0.5 indicate sparse vegetation or aging plants. Most of the hazelnut trees have NDVI values around 0.5, while a few in the northwest section have NDVI values higher than 0.5. NDVI values above 0.6 indicate dense vegetation at peak growth. The highest NDVI values in the orchard are those of the dense trees in the northwest, which are not hazelnut trees. The northeast and southeast sections of the orchard have higher NDVI values than the middle, where the grass had been recently mowed.

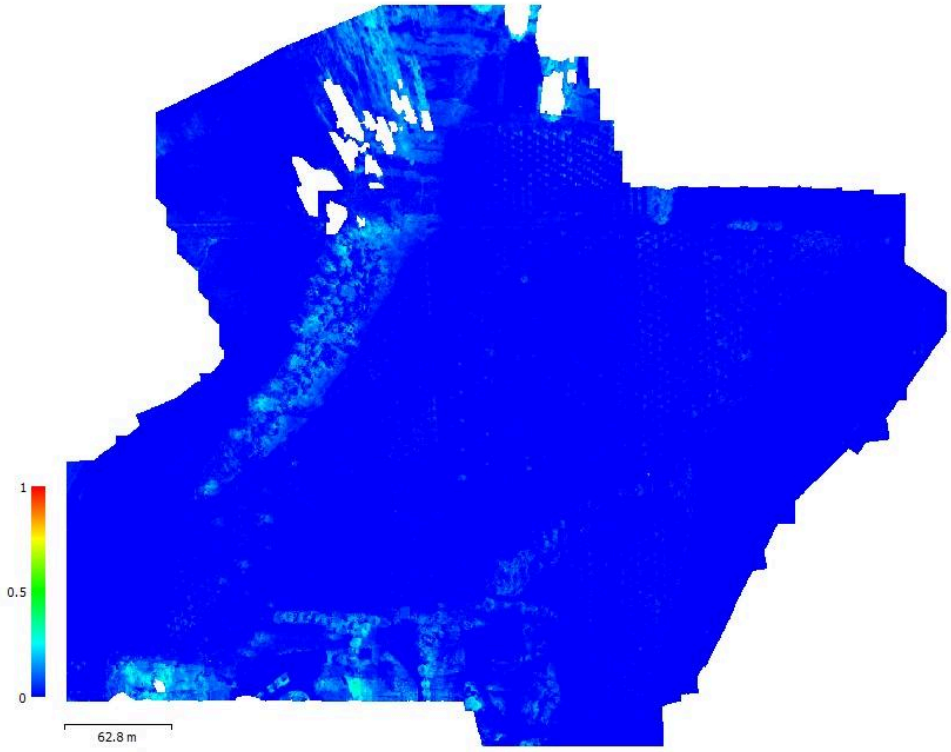


Figure 4: NDRE of the main hazelnut orchard created on Agisoft Metashape Pro.

NDRE values of 0.2 or less indicate bare soil or developing crops. NDRE values of 0.2 - 0.6 indicate unhealthy plants or plants that are not mature yet. NDRE values above 0.6 indicate healthy, mature, and ripening crops. All NDRE values in the hazelnut orchard are below 0.5; the highest values appear to be around 0.3.

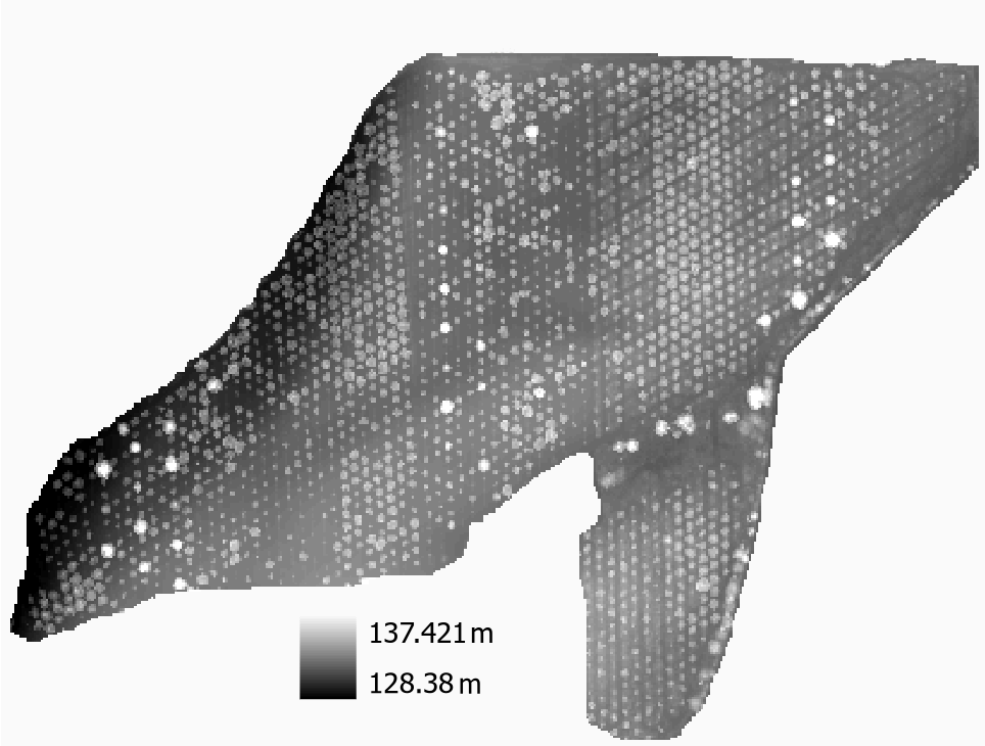


Figure 5: DSM of the main hazelnut orchard created in ArcGIS Pro.

Figure 5 shows the digital surface model (DSM) of the main hazelnut orchard. The DSM of the main orchard shows relatively low topographic variation across the study area, with elevations tightly ranging from a minimum of 128 meters to a maximum of 137 meters. The lowest elevation values, represented by the darker shades, are concentrated along the west side of the model. The highest elevation values, shown in the lighter shaded area, can be seen as circular points and show the taller tree crowns. The majority of these tall trees can be seen on the west and east ends of the orchard.

Discussion

A resolution of 5 cm/pixel allowed for differentiation between individual trees, which was necessary to answer the stated research questions. A resolution higher than 5 cm/pixel was not necessary, as we did not need to determine differences in individual leaves.

Hazelnut trees with higher NDVI values in the northwest section of the orchard might be healthier as a result of shading at certain times of the day. Shading of these trees by the taller trees of different species can be observed in the orthoimage. Additionally, hazelnut trees that are shaded appear to be slightly fuller than those toward the center of the orchard. Low NDVI values indicate soil throughout the orchard, as well as buildings in the southern part of the orchard. Areas of the orchard that were not mowed, in the northeast, show denser vegetation. This agrees with field observations of grasses a few feet tall.

None of the hazelnut trees in the orchard are in the mature or ripening stage according to results from the NDRE. This makes sense, given that the hazelnut harvest season in Oregon is in September and October. Higher NDRE values are observed in the southernmost part of the orchard and the northwest. These trees are not hazelnut species and likely have different growing seasons, leading to higher NDRE values. Ultimately, we determined that May is too early in the year to use an NDRE on hazelnut trees. The NDVI was not oversaturated, indicating that the NDRE might not be helpful at this time of year.

The higher NDVI values and fuller-looking canopy structures observed in the shaded northwest section of the orchard could be due to photosynthetic acclimation. When nut trees are subject to intermittent shading, they often undergo modifications to adjust for the lack of light. One possible interpretation is that the trees developed larger leaf canopies to capture more available light. This could inflate NDVI or NDRE values even if overall tree health is not necessarily higher. It appears that the trees in the shaded area changed their structure by growing extra leaves and spreading them out more widely. This change makes the trees look thicker compared to the trees in the unshaded areas. In addition, leaves in shaded areas may produce

more chlorophyll to make better use of the limited sunlight. This extra chlorophyll can make the leaves appear darker green compared to the rest of the trees in the orchard.

Finally, the DSM shows that while the majority of the orchard is fairly flat, there is a drop in elevation on the west side, suggesting that more water drainage may accumulate in that area. There doesn't appear to be a huge correlation between the low-elevation and high-NDVI areas, but we do see some trees with brighter values on that side. NDVI values are unfortunately pretty saturated on the north and southeast sides of the orchard, limiting our ability to visually compare that area to the rest of the orchard.

Limitations

We came across a few limitations in this study; the first issue was during the data collection – the LiDAR data had to be converted from DJI's proprietary format (.LDR) into.LAS files before we could perform any analysis. This required extra software and processing steps before the data was handed off to us. Another limitation was the high winds during the drone flight; strong winds can force the drones to tilt to maintain their flight, which could have altered the camera angle and added some distortion to the final product. Additionally, because the LiDAR and multispectral imagery were captured on different days and times, the shadow lengths based on differing sun positions and differences in orchard conditions may have altered some spectral analysis. Lastly, we felt that the time constraint was one of the biggest limitations that made it difficult for us to expand further on this analysis. Given more time, a few improvements we could have made are using the high-resolution imagery to detect more subtle changes, and adjusting some of the figure symbology to better show the differences.

Conclusion

Overall, our study found that partial shading may have influenced apparent hazelnut tree health and canopy structure at Farmhands Co-op, while topography and drainage appeared to have little relationship with our observed NDVI and NDRE patterns. The LiDAR data showed that the orchard was free from pooling and that the topography had little impact on the grove health. A soil analysis or ground survey would be best to further explore the drainage of the soil and the acidity level. As for our methods, we've concluded that the use of 5 cm/pixel resolution was detailed enough for this analysis since our focus was on individual trees.

The northwest section of the orchard was an interesting find, with the trees that experience more shade having fuller canopies than other hazelnut trees. The higher NDVI values observed in trees under higher rates of shade may have been due to photosynthetic acclimation. We also found that the NDVI was more useful than the NDRE in this project, likely due to May being too early for hazelnut season. Additional analyses like soil testing, harvest counts, and drone flights during optimal hazelnut tree season could yield more informative results.

Contributions

Tara Disser - Attended the field day, processed RGB, MS, and LiDAR data using Agisoft Metashape, ArcGIS Pro, and R (RStudio) to produce deliverables. Light editing of the report.
Mia Schuman - I attended the field day and was present for most of the data analysis (but of course, Tara was the one who actually processed the data!). I created most of the presentation and wrote the methods, results, and most of the discussion section of the paper.
Molly Cobbs - I was not able to attend the field day, but was there for the data analysis portion and helped out with some of the analysis as well as working on the presentation. Additionally, I

worked on the paper portion, primarily focusing on the project summary, conclusion, some of the limitations section, and editing.

Harrison Smith - I was not able to attend the field day, but was there for a little of the data analysis. I did some slides for the presentation and wrote the DSM analysis, limitations and some of the discussion.

Brandon Watkins - I was not able to attend the field day and wasn't able to be present for any of the data analysis. I did some of the slides and parts of the report. I unfortunately could not have been as present as I liked and my contributions weren't as much as others in the group.